

1a) $\{s1, s2\}$

b) $\{s1\}$

c) $\emptyset$

d) $\emptyset$

e) $\{s2\}$

f) $\{s0, s1, s2\}$

g) $\emptyset$

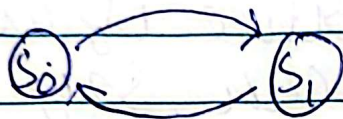
h) $\{s0, s1, s2\}$

2a) ~~AG~~ $\rightarrow$ AG (invalid-login $\rightarrow$
AG($\neg$ withdraw-money))

b) AG (open $\rightarrow$ A (open $\vee$ request-
to-money))

1) $AG(\text{request_site_update} \rightarrow AF(\text{updating_site} \wedge \neg AF(\text{update_complete})))$

3. Let us consider the following Kripke structure:



In this we have 2 states S_0 and S_1 , 2 transitions $S_0 \rightarrow S_1$ and $S_1 \rightarrow S_0$, which basically forms a cycle and let us consider that $L(S_0) = \{p\}$ and $L(S_1) = \{q\}$

Now first let us understand what the statements $AF(p \wedge q)$ and $AF(p) \wedge AF(q)$ mean:

- $AF(p \wedge q)$: On all paths, we will eventually hit a state where p and q will be true at the same time.
- $AF(p) \wedge AF(q)$: This means that on all paths, we will eventually hit a p -state and may be later, we

will hit a q -state as well.

Now, by looking at the example that we have used, we can say that it most likely ~~disproves~~ with the ~~stat~~ meaning of the statement $AF(p) \wedge AF(q)$, since if we argue at state s_0 , we can say that $AF(p)$ will be true in this case, since $p \in L(s_0)$, so p holds at step 0 and $AF(q)$ also holds at this state, since our next state is s_1 and $q \in L(s_1)$, also since its cyclic Kripke structure s_1 will eventually return to s_0 , $AF(p)$ also holds at s_1 and similarly $AF(q)$ will also be true at that since $q \in L(s_1)$, however, in this both p and q will not be true on the same state ever.

Hence ~~XXXXX~~ disproved.

4) $AG(p \wedge q)$: It means that on every path, at every state, p and q are true.

$AG(p \rightarrow AF(q))$: It means that in every state that's reachable, if p is true then along all paths we must eventually hit a q -state.

Now, if we look at the meanings of $AG(p \wedge q)$, it says that q is true at every state, so we can say that $AF(q)$ is true at every state and hence $p \rightarrow AF(q)$ will also turn out to be true since right side of the implication turns out to be true and that means that $AG(p \rightarrow AF(q))$ is also true.

Implication

Hence the ~~statements~~ $AG(p \wedge q)$ and $AG(p \rightarrow AF(q))$ turn out to be a tautology, which means that $AG(p \wedge q)$ will always be true whenever we find $AG(p \rightarrow AF(q))$.

5a) Claim: $A(p \circ \text{false}) \equiv A \neg(p)$
Therefore, it can be expressed in CTL.

a) Proof: $A(p \circ \text{false})$

~~Now, we~~ Option 1

- there exists a state where ~~ϕ_2~~ ^{false} is satisfied and before that ~~ϕ_1~~ holds in all states in all states in the path.

Option 2

- all states in the path satisfy p and not false

Now, by looking at these 2 options, we can say that ~~option 1~~ we will go with option 2, since for option 1 false must be true which is not possible and option 2 says that all states satisfies p which means it is true and is not false, which

makes more sense, hence it can be expressed in a CTL since it follows option 2.

CTL form: $AG(p)$ ("p holds forever")

b) P true: $AG \text{ false} \circ p$

Option 1

There exists a state where p is satisfied and before that p was false in all states in the path

Option 2

all states satisfy false and not p

By looking at the options, it seems like option 1 will satisfy our claim because it makes more sense since p is satisfied and before that p ^{was} false and it can be expressed as a CTL formula.

CTL formula ("p holds now"):